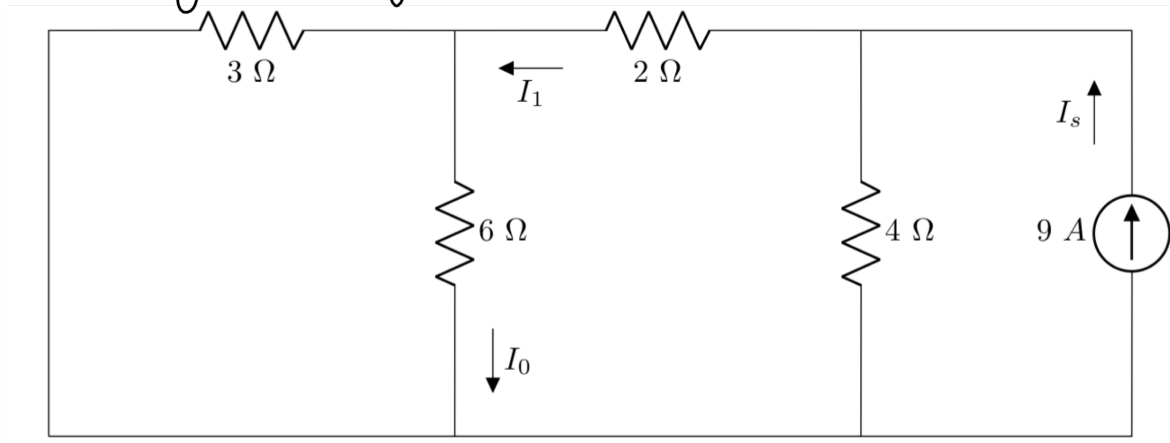


tutorial 5

4.4 Using linearity find I_o/I_s



Let $I_o = 1\text{ A}$, use current division

$$I_o = I_1 \times \frac{3}{3+6} = 1$$

$$\therefore I_1 = 3\text{ A}$$

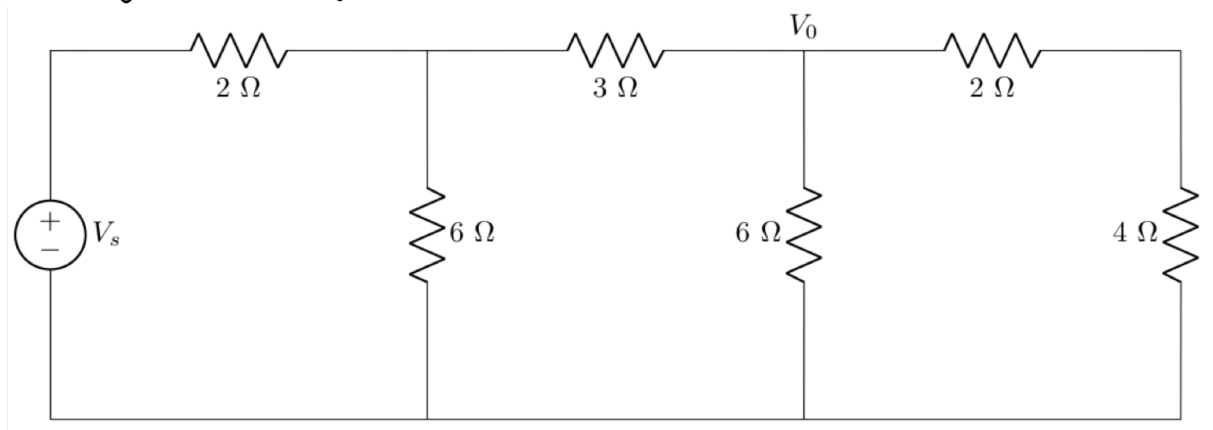
$$I_1 = I_s \times \frac{4}{3 \parallel 6 + 2 + 4} = 3$$

$$I_s \times \frac{4}{2+2+4} = 3$$

$$\therefore I_s = 6\text{ A}$$

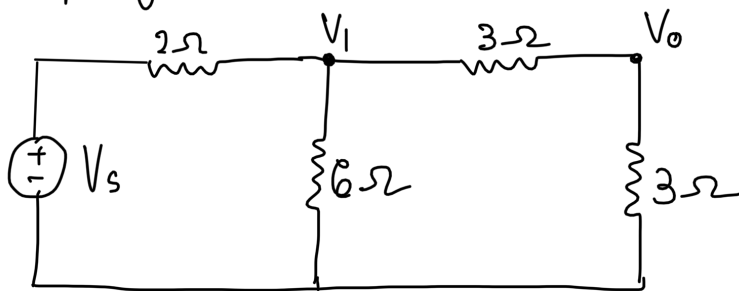
$$\therefore \frac{I_o}{I_s} = \frac{1}{6} \\ = 0.1667$$

4.5 Using linearity find V_s/V_o



Let $V_o = 1V$

⇒ Simplify the circuit

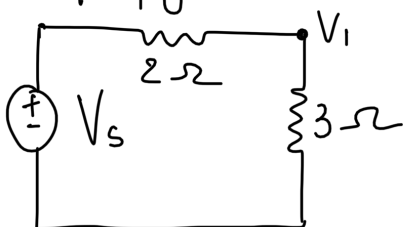


Use voltage division to find V_1

$$V_o = V_1 \times \frac{3}{3+3} = 1$$

$$\therefore V_1 = 2V$$

Simplify the circuit further

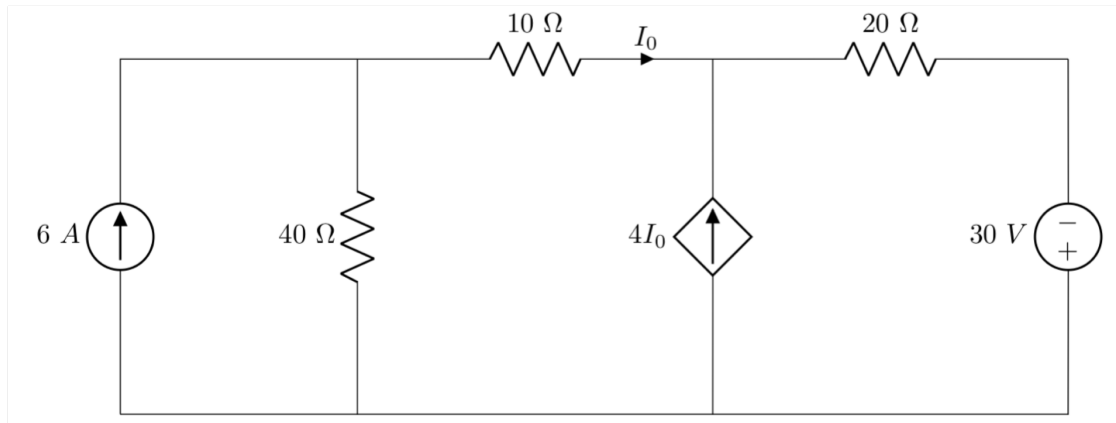


$$V_1 = V_s \times \frac{3}{2+3} = 2$$

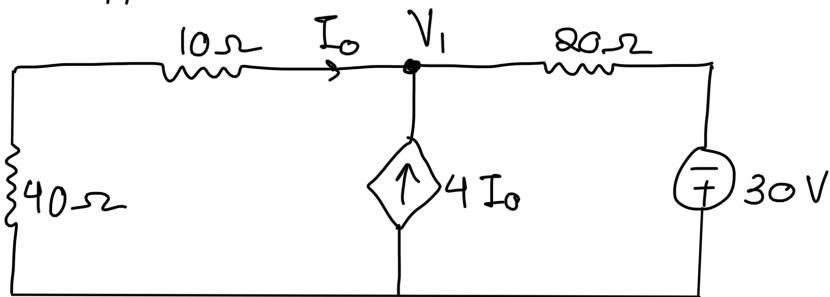
$$\begin{aligned}\therefore V_s &= \frac{10}{3} \\ &= 3.333 \text{ V}\end{aligned}$$

$$\begin{aligned}\therefore \frac{V_s}{V_0} &= \frac{3.333}{1} \\ &= 3.33\end{aligned}$$

- 4.11 (a) Find the contribution of the 30 V source to I_0
 (b) Find the contribution of the 6 A source to I_0



(a) Switch off current source



$$\frac{V_1}{10+40} - 4I_0 + \frac{V_1 - (-30)}{20} = 0$$

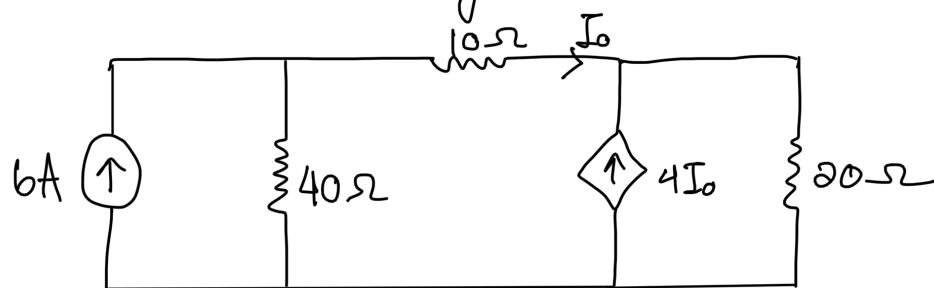
$$\text{where } I_0 = -\frac{V_1}{10+40} = -\frac{V_1}{50}$$

$$\therefore \frac{V_1}{50} + \frac{4V_1}{50} + \frac{V_1+30}{20} = 0$$

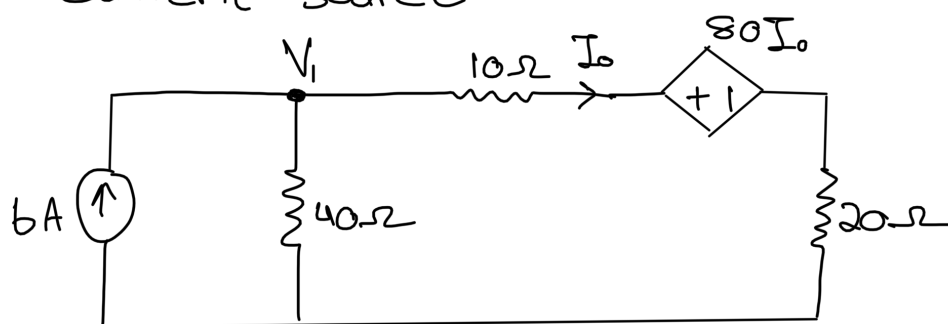
$$V_1 = -10 \text{ V}$$

$$I_0 = -\frac{(-10)}{50} = 0.2 \text{ A}$$

(b) Switch off voltage source



Do a source transform for the dependent current source



$$\frac{V_1}{40} - 6 + \frac{V_1 - 80I_0}{30} = 0$$

where $I_0 = \frac{V_1 - 80I_0}{30}$

$$V_1 = 30I_0 + 80I_0$$

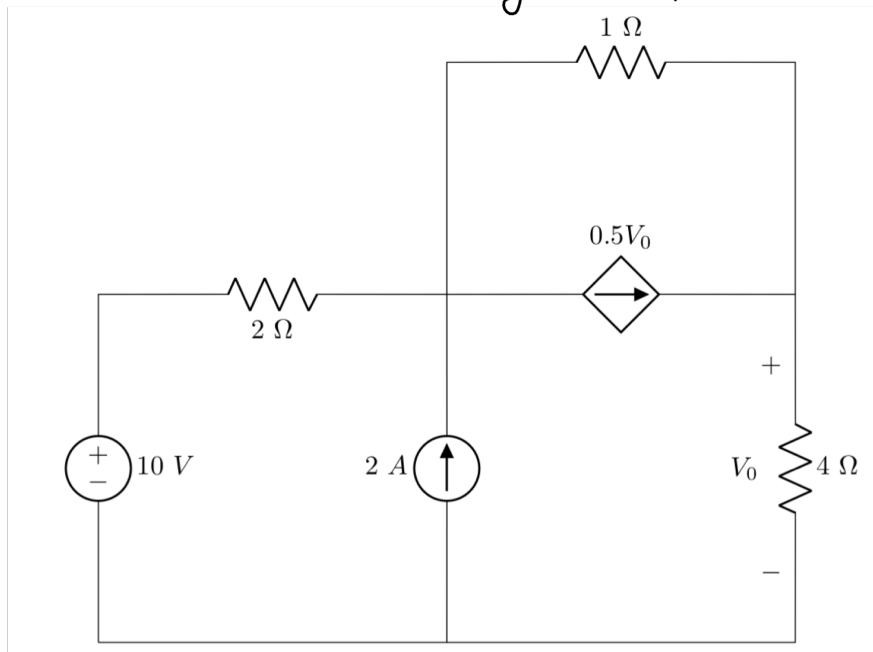
$$V_1 = 110I_0$$

$$\therefore \frac{110I_0}{40} + \frac{110I_0 - 80I_0}{30} = 6$$

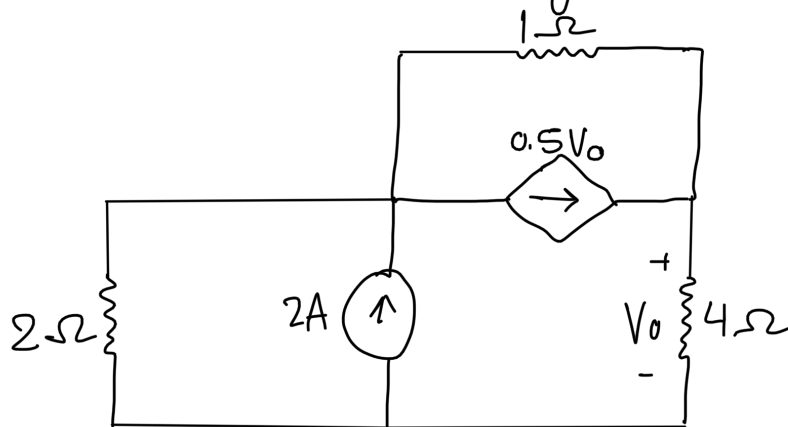
$$\frac{110I_0}{40} + \frac{30I_0}{30} = 6$$

$$I_0 = 1.6 \text{ A}$$

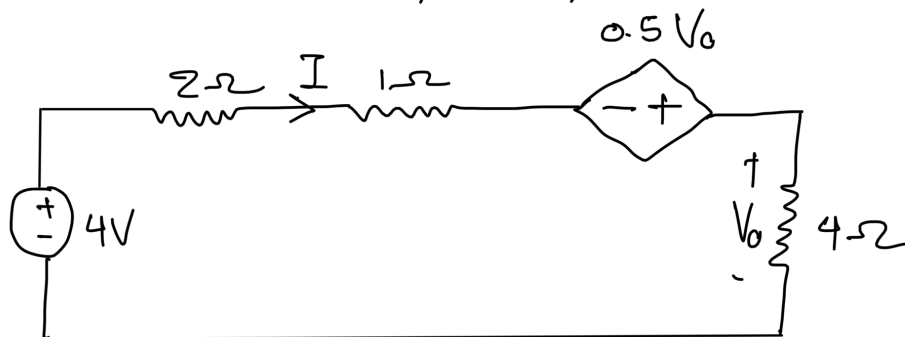
4.18 Determine V_0 using superposition



⇒ Switch off the 10V voltage source



⇒ Do a source transform for both sources



KVL :

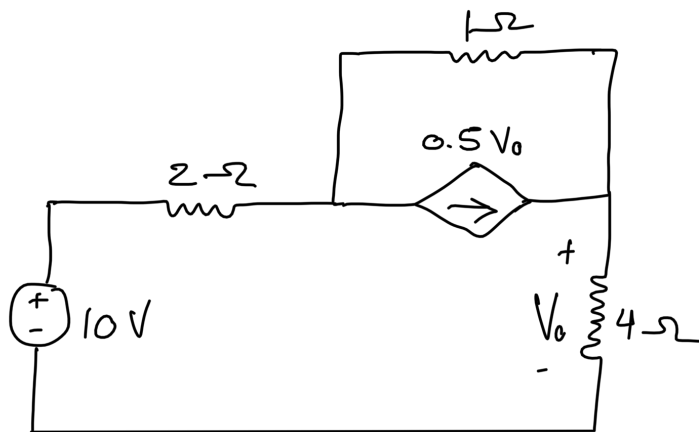
$$-4 + 3I - 0.5V_o + 4I = 0$$

where $V_o = 4I$

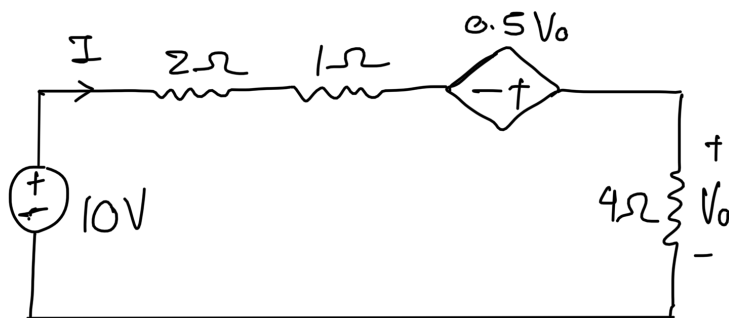
$$\therefore -4 + \frac{3V_o}{4} - 0.5V_o + V_o = 0$$

$$V_o = 3.2 \text{ V}$$

⇒ Switch off the 2A source



⇒ Source transform the 0.5V_o A dependent source



KVL :

$$-10 + 3I - 0.5V_0 + 4I = 0$$

where $V_0 = 4I$

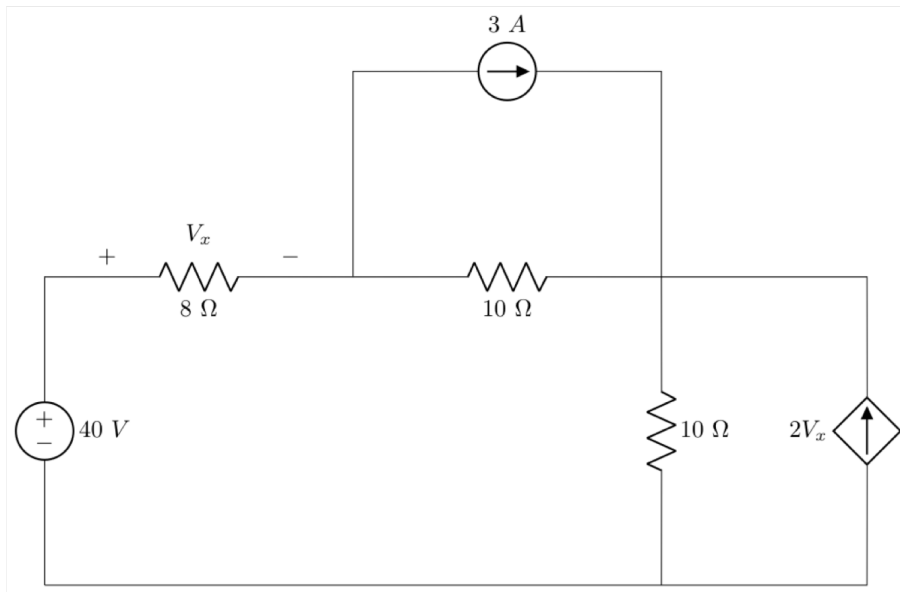
$$\therefore -10 + \frac{3V_0}{4} - 0.5V_0 + V_0 = 0$$

$$V_0 = 8V$$

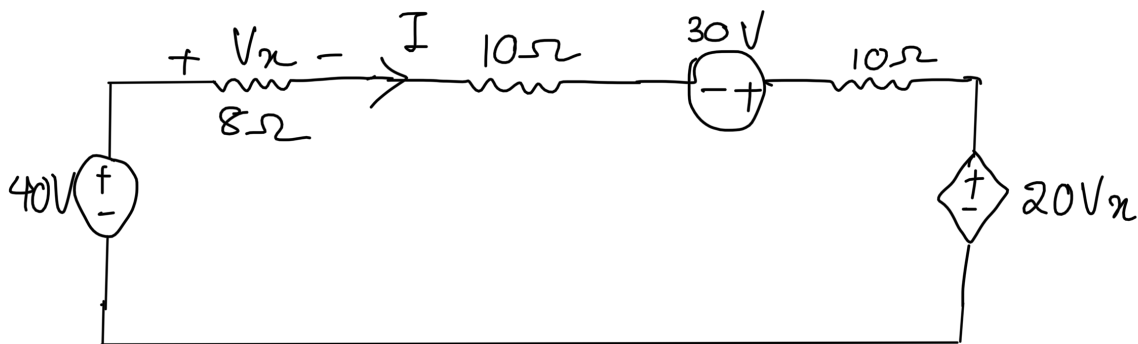
Add the two V_0 values :

$$\begin{aligned} V_0 &= 8 + 3.2 \\ &= 11.2 V \end{aligned}$$

4.24 Use source transformation to V_x



⇒ Source transform the 3A and $2V_x$ sources



$$V_x = 8I$$

KVL :

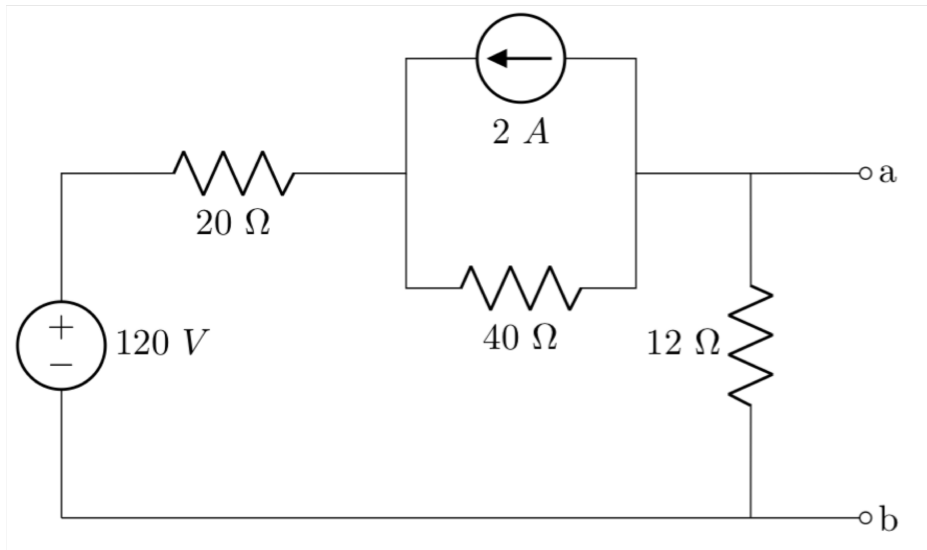
$$-40 + 8I + 10I + 10I - 30 + 20V_x = 0$$

$$28I + 20V_x = 70$$

$$\frac{28V_x}{8} + 20V_x = 70$$

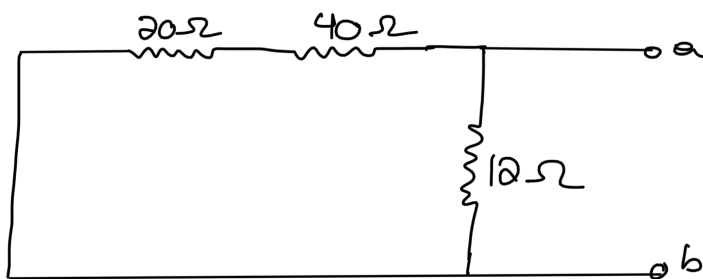
$$V_x = 2.979 \text{ V}$$

4.37 (a) Determine R_N
 (b) Determine I_N



(a) R_N

$\Rightarrow$ Switch off all independent sources

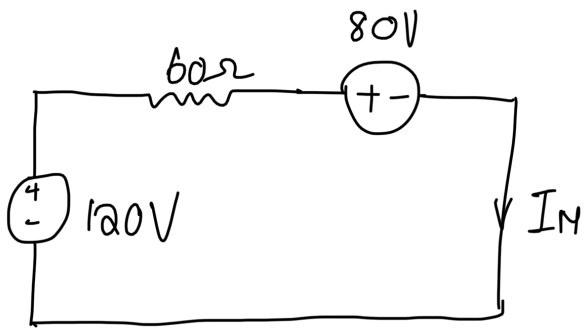
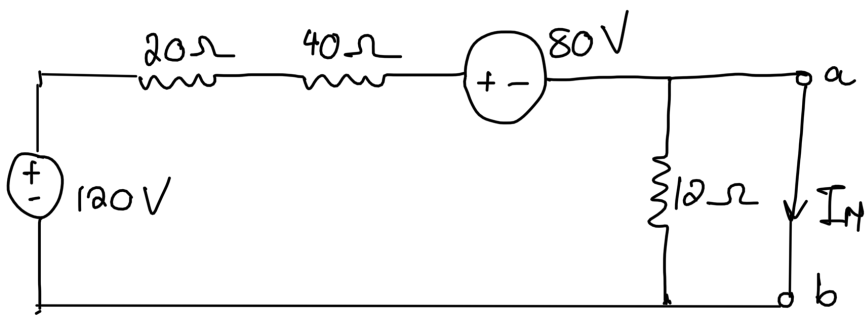


$$R_N = (20 + 40) \parallel 12$$

$$= 10 \Omega$$

(b) I_N

$\Rightarrow$ Source transform the 2A source and connect a to b with a short circuit

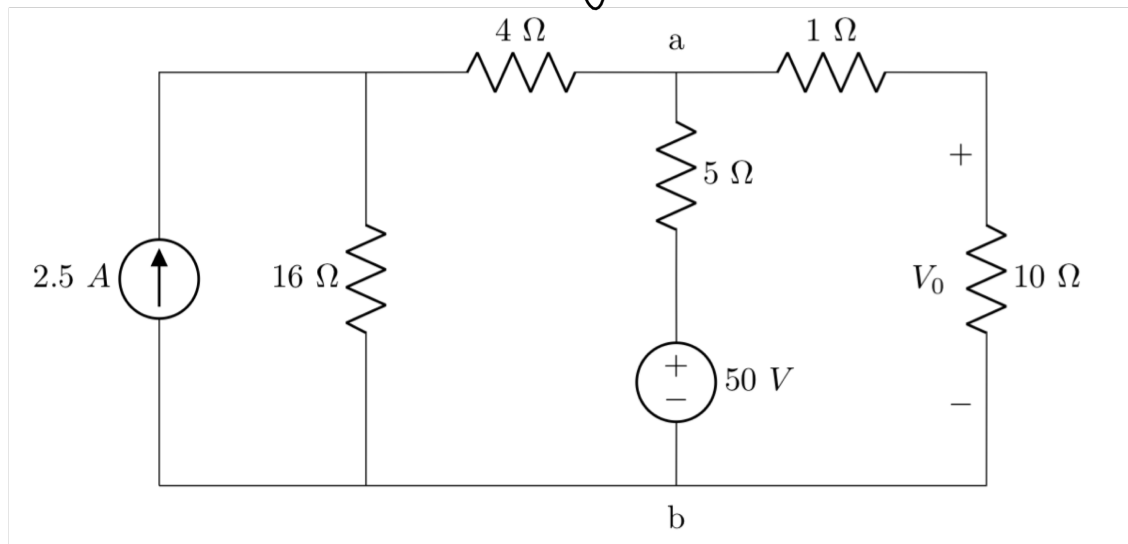


KVL :

$$-120 + 80 + 60I_N = 0$$

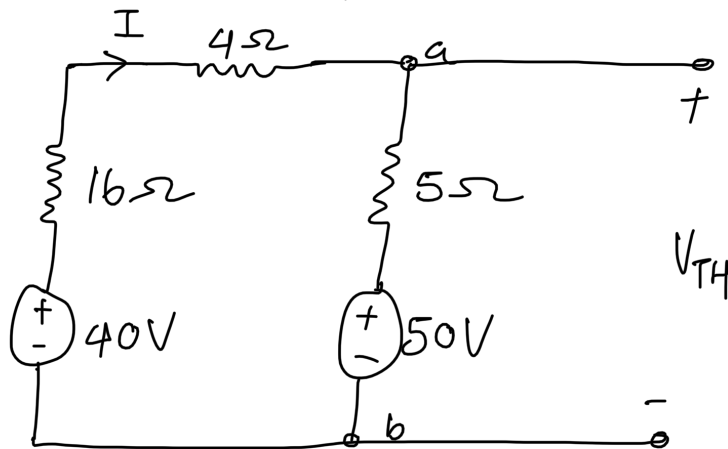
$$I_N = 0.667 \text{ A}$$

4.38 Determine V_o by finding V_{TH} and R_{TH} across a-b



⇒ Replace the 10Ω resistor with an open V_{TH} .
The 1Ω resistor is also removed since no current flows through it.

⇒ Source transform the 2.5 A source



KVL around loop :

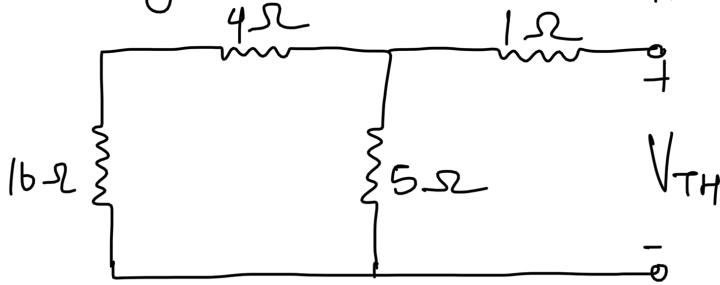
$$-40 + 16I + 4I + 5I + 50 = 0$$

$$25I = -10$$

$$I = -0.4\text{ A}$$

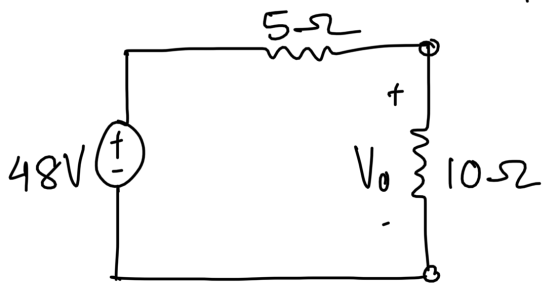
$$\begin{aligned}
 V_{TH} &= 5I + 50 \\
 &= 5(-0.4) + 50 \\
 &= 48 \text{ V}
 \end{aligned}$$

⇒ Finding R_{TH} . Switch off all sources:



$$\begin{aligned}
 R_{TH} &= (16 + 4) \parallel 5 + 1 \\
 &= \frac{20(5)}{25} + 1 \\
 &= 5\Omega
 \end{aligned}$$

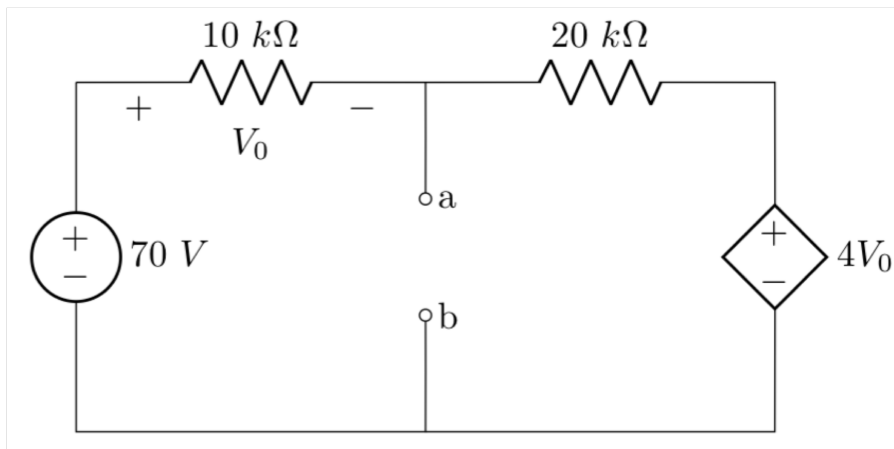
⇒ Draw the Thevenin equivalent circuit



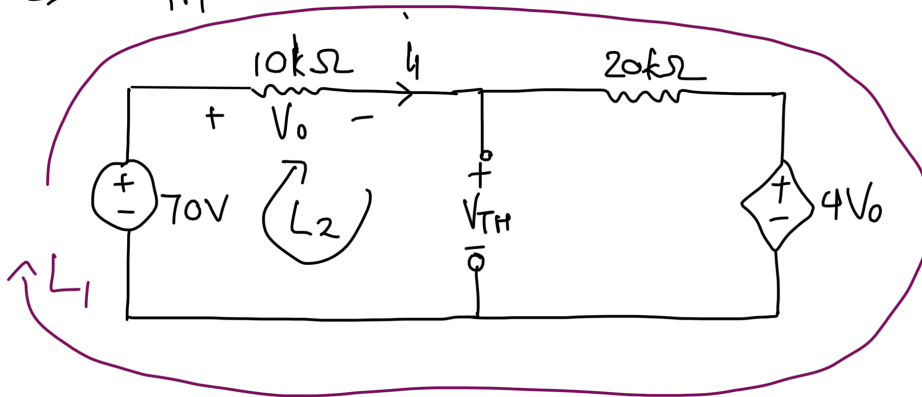
Voltage division :

$$\begin{aligned}
 V_0 &= 48 \times \frac{10}{10 + 5} \\
 &= 32 \text{ V}
 \end{aligned}$$

4.40 Determine V_{TH} and R_{TH} across a-b



$\Rightarrow V_{TH}$



$$L_1: -70 + 10000 i_1 + 20000 i_1 + 4V_0 = 0$$

where

$$V_0 = 10000 i_1$$

$$\therefore 30000 i_1 + 4(10000 i_1) = 70$$

$$i_1 = 1 \text{ mA}$$

$$V_0 = 10000 (1 \times 10^{-3})$$

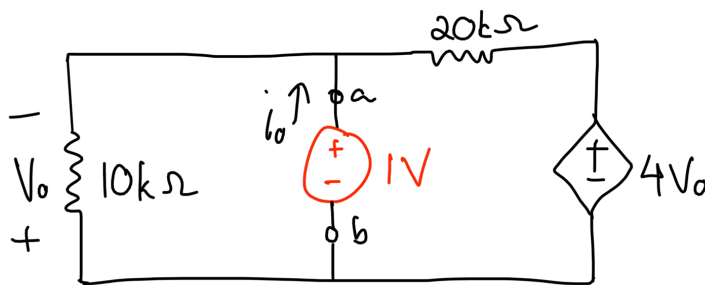
$$= 10 \text{ V}$$

$$L_2 : -70 + V_o + V_{TH} = 0$$

$$\begin{aligned} V_{TH} &= 70 - V_o \\ &= 70 - 10 \\ &= 60 \text{ V} \end{aligned}$$

⇒ R_{TH}

- ① Switch off the dependent sources
- ② Add test source across the a-b terminals
- ③ Solve for the current or voltage across the test source



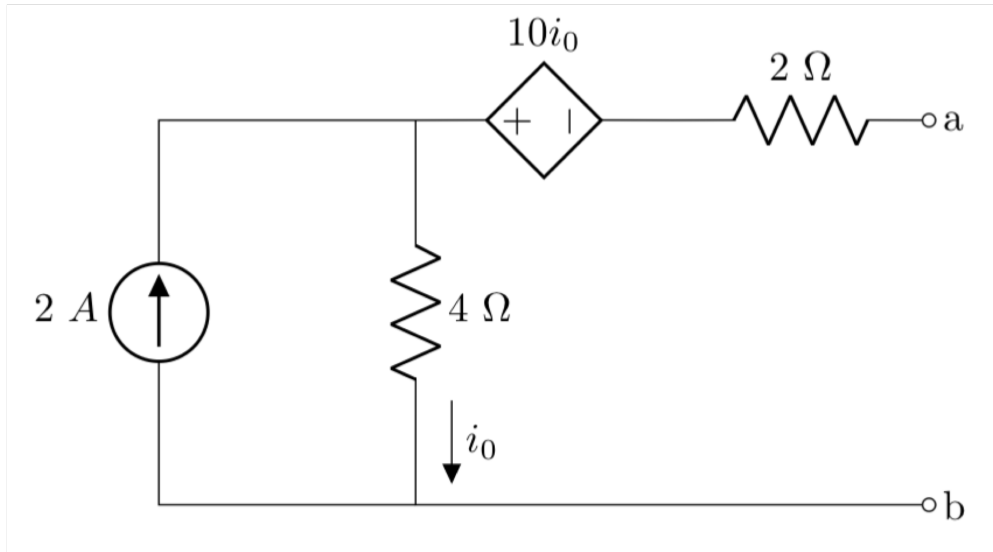
KCL :

$$-i_o + \frac{1}{10000} + \frac{1 - 4(-1)}{20000} = 0$$

$$i_o = 350 \mu\text{A}$$

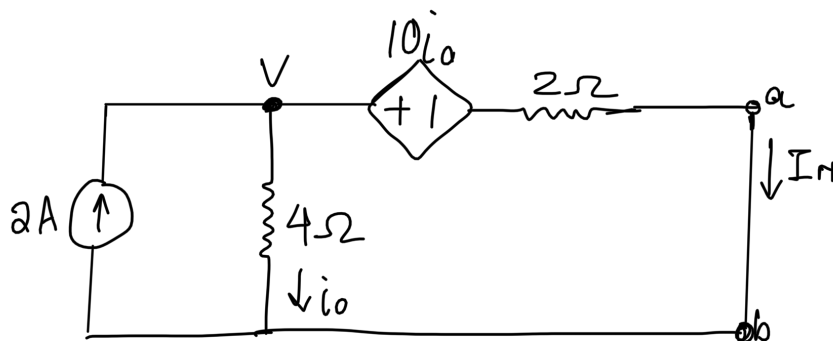
$$\begin{aligned} \therefore R_{TH} &= \frac{V_{\text{TEST}}}{i_o} \\ &= \frac{1}{350 \times 10^{-6}} \\ &= 2857 \Omega \end{aligned}$$

4.48 Determine I_N and R_N across a-b



⇒ I_N

↪ Connect terminals a-b with a short circuit



KCL:

$$-2 + \frac{V}{4} + \frac{V - 10i_0}{2} = 0$$

where

$$V = 4i_0$$

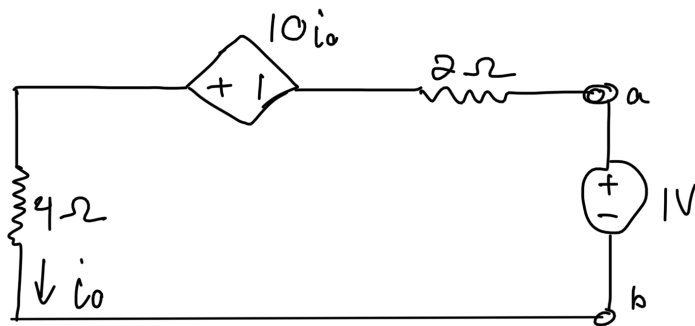
$$\therefore -2 + \frac{V}{4} + \frac{V - 10V/4}{2} = 0$$

$$V = -4V$$

$$\begin{aligned}
 I_N &= \frac{V - 10i_o}{2} \\
 &= \frac{V - 10V/4}{2} \\
 &= \frac{-4 - \frac{10(-4)}{4}}{2} \\
 &= 3 \text{ A}
 \end{aligned}$$

$\Rightarrow R_N$

$\leadsto$ Switch off current source and add a 1V test source between terminals a-b



KVL:

$$4i_o - 1 + 2i_o - 10i_o = 0$$

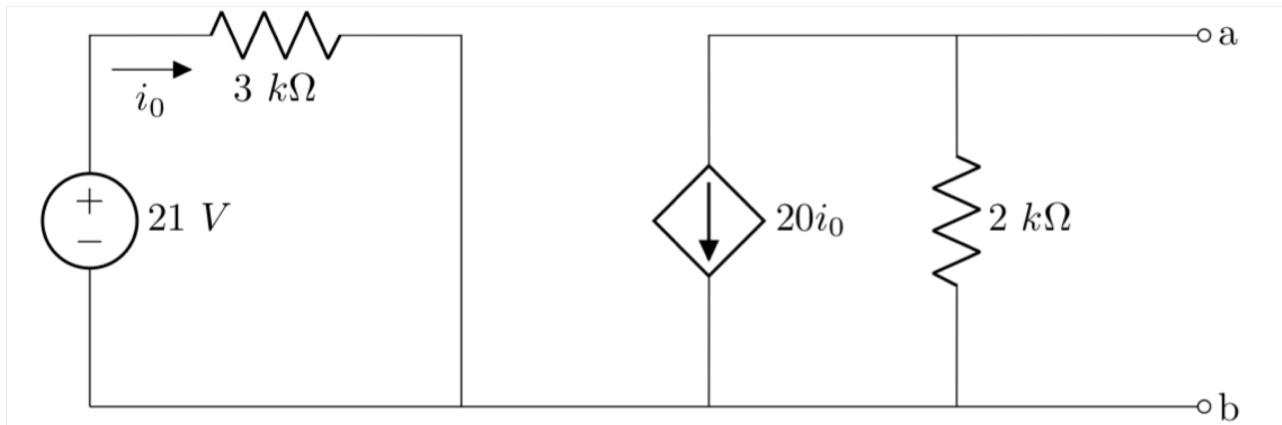
$$-4i_o = 1$$

$$i_o = -0.25 \text{ A}$$

$$\begin{aligned}
 R_N &= \frac{1}{0.25} \\
 &= 4 \Omega
 \end{aligned}$$

$\leftarrow$ Note: current and voltage directions are ignored because resistance cannot be negative

4.5a Determine V_{TH} and R_{TH} across a-b



$\Rightarrow V_{TH}$

$$i_0 = \frac{21}{3000} \\ = 7 \text{ mA}$$

$$V_{2k\Omega} = 2000 (20 i_0) \\ = 2000 (20) (7 \times 10^{-3}) \\ = 280 \text{ V}$$

check direction

$$\therefore V_{TH} = -280 \text{ V}$$

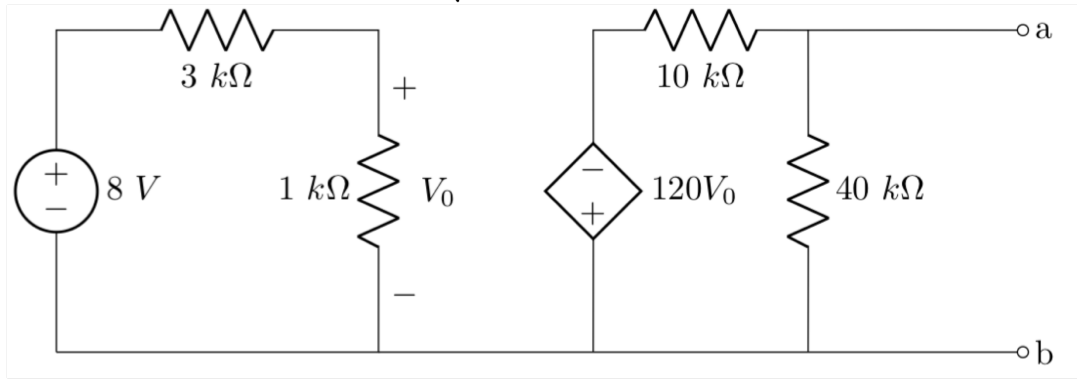
$\Rightarrow R_{TH}$

Switch off independent source.

$$\therefore i_0 = 0 \text{ A}$$

$$R_{TH} = 2 \text{ k}\Omega$$

4.71 Determine $P_{\max}$ for a load across a-b



$\Rightarrow V_{TH}$

$\leadsto$ Voltage division

$$V_0 = 8 \times \frac{1000}{1000 + 3000}$$
$$= 2V$$

$$V_{TH} = V_{40k\Omega}$$

$$= -120 V_0 \times \frac{40000}{40000 + 10000}$$

$$= -120(2) \times \frac{4}{5}$$

$$= -192V$$

$\Rightarrow R_{TH}$

① Switch off the independent source

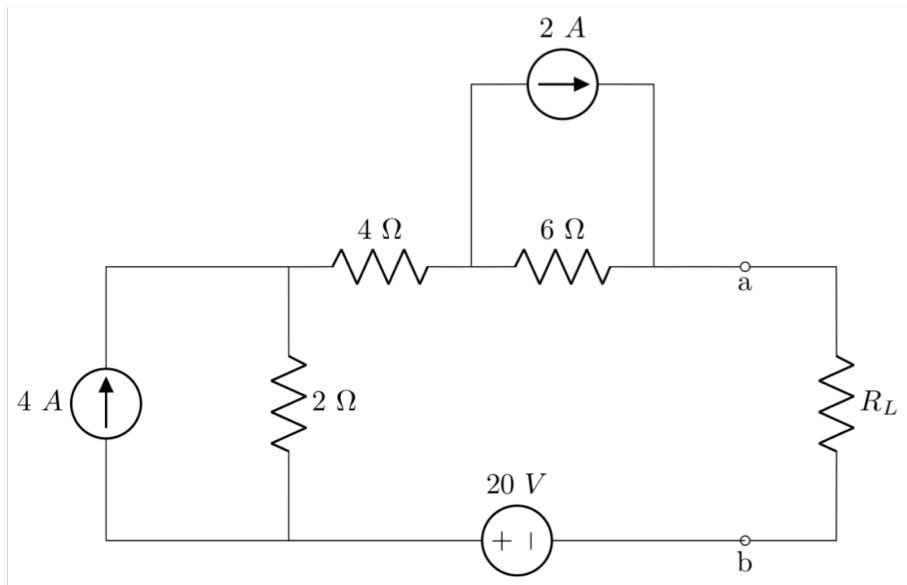
② This means $V_0 = 0V$ and the dependent source is also off

$\therefore$ No need for test voltage

$$\begin{aligned}
 \therefore R_{TH} &= 10\,000 \parallel 40\,000 \\
 &= \frac{(10\,000)(40\,000)}{10\,000 + 40\,000} \\
 &= 8\,k\Omega
 \end{aligned}$$

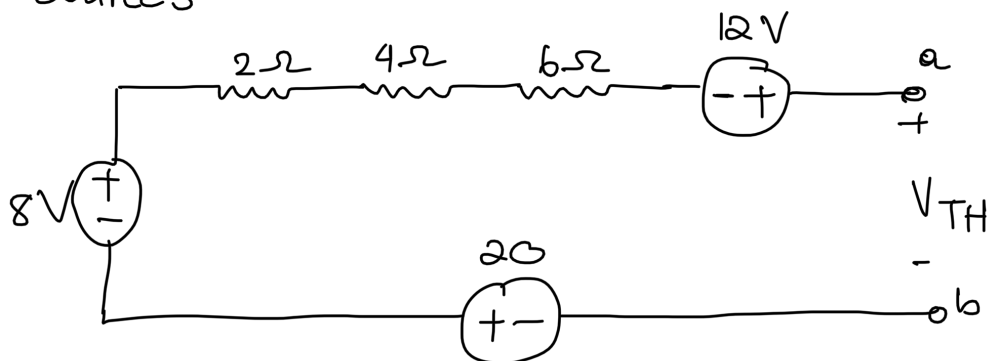
$$\begin{aligned}
 P_{max} &= \frac{V_{TH}^2}{4R_{TH}} \\
 &= \frac{(-192)^2}{4(8000)} \\
 &= 1.152\,W
 \end{aligned}$$

4.7a Determine the current delivered to the load resistor by finding the Thevenin equivalent across a-b.
 $R_L = 8\ \Omega$



$\Rightarrow V_{TH}$

Perform source transformation for the 2A and 4A sources



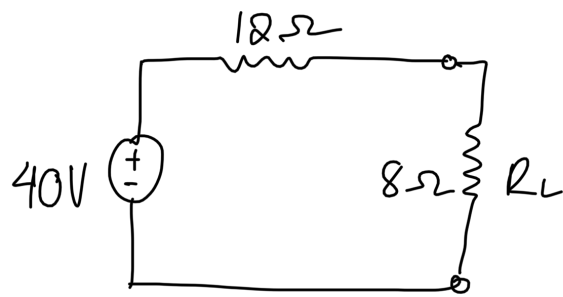
$$-8 - 12 + V_{TH} - 20 = 0$$

$$V_{TH} = 40\text{ V}$$

$\Rightarrow R_{TH}$

Switch off all independent sources

$$R_{TH} = 2 + 4 + 6 \\ = 12\ \Omega$$



$$\therefore \bar{I}_{RL} = \frac{40}{(8+12)} \\ = 2A$$